

UNDERACTUATED ROBOTICS

Algorithms for Walking, Running, Swimming, Flying, and Manipulation

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Note: These are working notes used for [a course being taught at MIT](#). They will be updated throughout the Spring 2024 semester. [Lecture videos are available on YouTube](#).

[Previous Chapter](#)

[Table of contents](#)

[Next Chapter](#)

APPENDIX D

An Optimization Playbook

This chapter is intended to provide a collection of tips and formulations that can make seemingly non-smooth constraints smooth, seemingly non-convex constraints convex, etc. (It's very much a work in progress...)

D.1 MATRICES

D.2 ELLIPSOIDS

Imagine an ellipsoid centered at the origin: $E = \{x | x^T S x \leq 1\}$ with $S = S^T \succ 0$. The volume of this ellipse is proportional to $(\det S^{-1})^{\frac{1}{2}}$. We know that $\log \det(S)$ is a concave function in the elements of S [1].

maximizing volume.

minimizing volume.

volume of a semi-algebraic set. (via a contained ellipsoid)

D.3 POLYTOPES

D.4 PERSPECTIVE FUNCTIONS

D.5 (MIXED-)INTEGER PROGRAMMING

D.6 BILINEAR MATRIX INEQUALITIES (BMIs)

D.7 GEOMETRY (SE(3), PENETRATION, AND CONTACT)

REFERENCES



1. Stephen Boyd and Lieven Vandenberghe, "Convex Optimization", Cambridge University Press , 2004.

[Previous Chapter](#)

[Table of contents](#)

[Next Chapter](#)

[Accessibility](#)

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